

Cloud Run, compressing prototype-to-production cycles through automated quality gates.

• Dry Ground AI

United States of America (Remote)

AI Engineer

Mar. 2025 – Nov. 2025

- **AI Voice Agent Development:** Developed and deployed AI voice agents to answer real estate-related calls; integrated tools to retrieve up-to-date information, send messages and emails, and automate client interactions.
- **Client Profiling & Lead Generation:** Developed a data science workflow for classifying user profiles and identifying potential clients, supporting strategic sales initiatives.

• WEG

Jaragua do Sul, Brazil

Data Scientist

Jun. 2024 – Mar. 2025

- **Predictive Maintenance & Anomaly Detection:** Trained an Isolation Forest model to detect anomalies in wind turbine converters, deployed as an API on AWS SageMaker; the model anticipated historical failures by up to two weeks and identified approximately 70% of the interventions listed in maintenance service orders.
- **Explainability & Root-Cause Diagnosis:** Developed a companion SHAP-based API enabling maintenance engineers to diagnose the root causes of detected anomalies, bridging automated detection and corrective action.
- **End-to-End MLOps:** Built a reproducible MLOps pipeline (MLflow, Docker, Kubernetes, AWS) spanning automated training, versioning, deployment, and live monitoring via Grafana dashboards backed by InfluxDB, enabling iterative model improvements at production scale with full experiment lineage tracking.

• Macnica DHW

Florianópolis, Brazil

Data Scientist

Nov. 2022 – Oct. 2023

- **Smart Sensors:** Developed an SVM-based classification pipeline for vibration sensor data; applied Multi-objective Genetic Algorithms (NSGA-II via `pymoo`) to optimize the trade-off between model accuracy and computational efficiency for edge hardware deployment.
- **Computer Vision APIs:** Built a dual-layer video anonymization API using OpenCV and DBSCAN clustering for motion-based person detection, with a pre-trained ResNet-50-FPN model (PyTorch) as a verification layer; optimized specifically for low frame-rate surveillance footage.
- **IoT Development:** Led end-to-end IoT product development with full Docker containerization of the edge-to-cloud stack (MQTT Broker, PostgreSQL, InfluxDB, Grafana, Python services); implemented edge computing including FFT calculations directly on Senspider hardware.

SSE Gridtech

Curitiba, Brazil

• Internship

Oct. 2014 – Dec. 2015

R&D Engineer

Jan. 2016 – Aug. 2016

- **Smart Grid & IoT Solutions:** Developed a long-range LoRaWAN-based communication system to optimize smart metering infrastructure, reducing GSM dependency.

Research Experience

• University of Warwick, Warwick Business School

Coventry, United Kingdom

Research

Oct. 2019 – Apr. 2022

- **Field of Research:** Ranking & Selection; Bayesian Optimization; Reinforcement Learning.
- **Research:** Investigated sample-efficient policy learning for automating constrained supply chain scheduling of biopharmaceutical products (personalized therapies); developed Multi-Objective MCTS (MO-MCTS) using marginal hypervolume as the selection criterion and integrated Evolutionary Algorithms to evolve base policies, combining Bayesian optimization with ranking-and-selection theory.
- **Collaboration:** Under supervision of Prof. Juergen Branke, we collaborated with Prof. Chun-Hung Chen on two-stage Ranking and Selection (R&S) problems, which contributed to research extending Optimal Computing Budget Allocation (OCBA) and Expected Value of Information (EVI) algorithms to efficiently allocate finite stochastic simulation budgets on Bayesian approaches for optimal decision-making.

• Shinshu University

Nagano, Japan

MSc Research

Apr. 2017 – Mar. 2019

- **Field of Research:** Multi-objective Optimization; Genetic algorithms; Evolutionary Computation.
- **Research:** Studied the impact of Pareto set topologies on multi-objective evolutionary algorithms (MOEAs) and developed a novel method to enhance their performance.
- **International Collaboration:** Worked with researchers from Japan, Mexico, and France.
- **Recognition:** Received the Student Best Paper Award at EMO-2019 (Michigan State University, US).

Education

• Shinshu University

Nagano, Japan

Master of Engineering in Electronics and Information System

Apr. 2017 – Mar. 2019

- MEXT Monbukagakusho Scholarship (Japanese Government)

• Concordia University

Montreal, Canada

Exchange Student, Electrical Engineering

Aug. 2012 – Aug. 2013

- Science Without Borders Scholarship (Brazilian Government)

• Federal University of Technology - Parana

Curitiba, Brazil

Bachelor of Engineering in Electronics

Jan. 2010 – Jul. 2016

Technical Expertise

Programming Languages:	Python, C, C++, Bash (Unix Shell).
Frontend Development:	React, JavaScript, TypeScript, HTML, CSS.
AI & Machine Learning Frameworks:	PyTorch, Hugging Face Transformers, scikit-learn, XGBoost, LangChain, LangGraph, Google ADK.
MLOps & Deployment Tools:	Docker, Kubernetes, MLflow, Git, CI/CD, FastAPI, Grafana.
Embedded Systems:	ARM Cortex-M, Jetson Nano, Arduino, FreeRTOS, MQTT.
Data Engineering & Pipelines:	PostgreSQL, Redshift, InfluxDB.
Cloud Platforms:	AWS (SageMaker, ECR, S3, EC2, Lambda), GCP (Cloud Run, Vertex AI).
Optimization:	Bayesian Optimization, Multi-objective Evolutionary Algorithms.
Operating Systems:	Linux (Fedora, Ubuntu), Windows.

Publications

- **Y. Marca**, H. Aguirre, S. Zapotecas, A. Liefoghe, B. Derbel, S. Verel, and K. Tanaka. Approximating Pareto set topology by cubic interpolation on bi-objective problems. 10th International Conference on Evolutionary Multi-Criterion Optimization (EMO 2019), Lecture Notes in Computer Science (LNCS), Michigan, USA. (**Best Student Paper Award**)
- **Y. Marca**, H. Aguirre, S. Zapotecas, A. Liefoghe, B. Derbel, S. Verel, and K. Tanaka. NSGA-II with Spline Interpolation on Bi-objective Problems with Difficult Pareto Set Topology. JPNSEC 2018 Symposium on Evolutionary Computation, Fukuoka, 2018. (**Young Research Award**)
- **Y. Marca**, H. Aguirre, S. Zapotecas, A. Liefoghe, B. Derbel, S. Verel, and K. Tanaka. MOEAs on Problems with Difficult Pareto Set Topologies. IEICE Shin-etsu Branch IEEE Session, Niigata University, 2018, p. 169. (**Young Research Award**)
- **Y. Marca**, H. Aguirre, S. Zapotecas, A. Liefoghe, B. Derbel, S. Verel, and K. Tanaka. Pareto dominance-based MOEAs on problems with difficult pareto set topologies. In Proceedings of the Genetic and Evolutionary Computation Conference Companion (GECCO '18). ACM, New York, NY, USA, 189-190.
- C. E. A. L. Rocha, **Y. P. Marca**, F. K. Schneider. Support Platform for Decision-Making in Research and Technological Development in Public Health. ESPACIOS (CARACAS), v. 39, p. 14-26, 2018.
- **Y. P. Marca**, S. Scholze. Proposta de Substituição da Comunicação GSM em Smart Grids por Rádios de Longo Alcance. XXXIII Simpósio Brasileiro de Telecomunicações, 2015, Juiz de Fora, MG. Anais Completo da Programação Técnica, 2015.
- **Y. P. Marca**, C. E. A. L. Rocha, B. Schneider Jr, F. K. Schneider. Plataforma de Apoio ao Processo Decisório em Pesquisa e Desenvolvimento Tecnológico em Saúde. Congresso Brasileiro de Engenharia Biomédica, 2012, Porto de Galinhas. ANAIS - CBEB 2012, 2012.

- M. P. Krause, D. M. Nakato, **Y. P. Marca**, F. K. Schneider. Gerenciamento do Controle da Glicemia Utilizando um Aplicativo para Celular. Congresso Brasileiro de Engenharia Biomédica, 2012, Porto de Galinhas. ANAIS - CBEB 2012, 2012.

Languages

Portuguese: Native

English: Fluent

Japanese: Conversational

Projects

- **End-to-End Machine Learning Pipeline for Short-Term Rental Price Prediction**
 - <https://github.com/yurimarca/build-ml-pipeline-for-short-term-rental-prices>
 - Developed a reproducible ML pipeline to predict short-term rental prices in New York City, ensuring scalability for weekly data updates.
 - Implemented data ingestion, cleaning, validation, and feature engineering processes to prepare datasets for modeling.
 - Trained and optimized a Random Forest regression model, employing hyperparameter tuning to enhance predictive performance.
 - Utilized MLflow for experiment tracking and model management, and integrated Weights & Biases for artifact tracking and visualization.
 - Designed the pipeline for seamless retraining with new data, facilitating continuous model improvement and deployment.